
Initialization DNA for a Small GPT Model: Structured Weight Geometry and Persistent Training Acceleration

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Abstract

1 Initialization is usually treated as a neutral prelude to optimization. We instead test
2 whether a compact, structured transformation of sampled parameters can encode
3 a reusable training advantage. We call the transformation an *initialization DNA*:
4 a form of prior information structure encoded as an executable list of parameter
5 groups, operations, layer scopes, and scalar values applied once before ordinary
6 training. Controlled paired experiments show that a selected DNA consistently
7 accelerates convergence while improving both early and fixed-budget behavior.
8 Projected weight geometry, log-count dynamics, component ablations, and paired
9 convergence trajectories support a working interpretation: initial weight distribu-
10 tions are not neutral, and sparse DNA edits accelerate movement toward a training-
11 favorable distribution. The experiments establish this claim only for the studied
12 model and do not assert cross-model transfer or a uniquely correct weight distri-
13 bution.

14 1 Introduction

15 Initialization controls the starting geometry from which an optimizer constructs a trained network.
16 Analytical and architecture-specific schemes have long shown that parameter scales alter signal prop-
17 agation and optimization [Ba et al., 2016, Huang et al., 2020, Zhang et al., 2019]. Yet standard prac-
18 tice often treats any valid random draw from the prescribed initializer as equivalent. We examine a
19 different empirical premise: useful information may be expressed as a sparse transformation of the
20 initial weight distribution, before gradients are observed.

21 An *initialization DNA* is a short program applied once to sampled parameters. It does not change
22 the architecture, objective, data order, optimizer, or training budget. This makes DNA a controlled
23 perturbation of the starting state rather than a replacement for gradient training. We study one
24 compact GPT throughout and reconstruct its recorded mutation-selection lineage from an empty
25 parent to a single LayerNorm gene and then a selected three-gene artifact.

26 The evidence is organized to separate selection from estimation. A targeted replay verifies the first
27 historical random proposal on disjoint seeds. The frozen single- and three-gene artifacts are then
28 compared with standard initialization over 30 paired seeds, six learning rates, and 20 paired 5000-
29 step runs. The three-gene artifact is the primary intervention; the single gene documents the first
30 discovery edge and acts as a component baseline.

31 **Related work.** Layer normalization and structured initialization demonstrate that scale is part of
32 model design rather than an incidental constant [Ba et al., 2016, Huang et al., 2020]. Maximal-
33 update parameterization studies stable hyperparameter transfer [Yang et al., 2022]; here dimensions

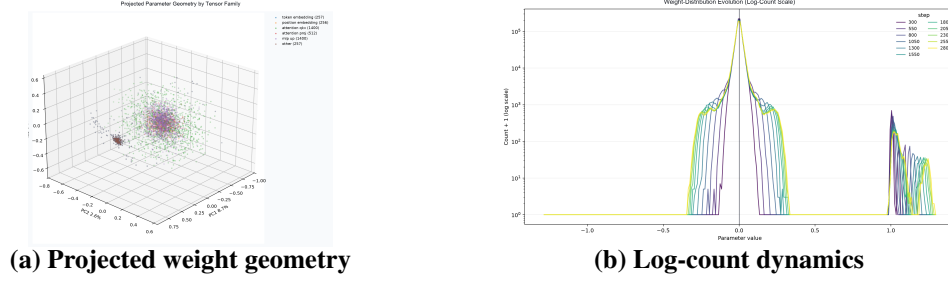


Figure 1: Motivating observations. **(a)** Parameter-family geometry suggests that weight distributions are non-neutral and may contain an information-bearing structure analogous to DNA. **(b)** The DNA-injected trajectory reorganizes during training. We interpret this, together with Figure 3, as accelerated movement toward a training-favorable distribution, not as causal identification of a unique correct distribution.

are fixed and learning rate is varied as a competing intervention. Evolutionary methods can optimize policies or weights [Such et al., 2017], whereas our mutations propose only a compact initialization program followed by unchanged causal language-model training.

2 Initialization DNA and Study Protocol

Let $\theta_0 \sim P$ be parameters sampled by the standard initializer. A DNA is an ordered list $D = (g_1, \dots, g_k)$ with

$$g = (\text{target}, \text{operation}, \text{scope}, \text{value}), \quad \tilde{\theta}_0 = T_D(\theta_0). \quad (1)$$

A scale gene maps each tensor selected by its target and scope from W to vW . After applying T_D , training is identical to the standard run. Paired groups share the same seed, sampled base initialization, data order, dropout randomness, optimizer state, and budget.

The study model is a three-block, four-head decoder-only GPT with width 148, context 256, byte tokenization, dropout 0.1, and batch size 64. The fixed default optimizer environment uses learning rate 3×10^{-4} , minimum rate 6×10^{-5} , five warmup steps, cosine decay over 2800 steps, weight decay 0.1, and gradient clipping at 1.0. The corpus combines 10.80 MB of Wikipedia-derived text and 1.12 MB of Shakespeare text, corresponding to approximately 11.9 million raw byte-level tokens.

The recorded lineage contains three ordered scale genes; Figure 2 pairs the executable seed with their anatomical intervention sites.

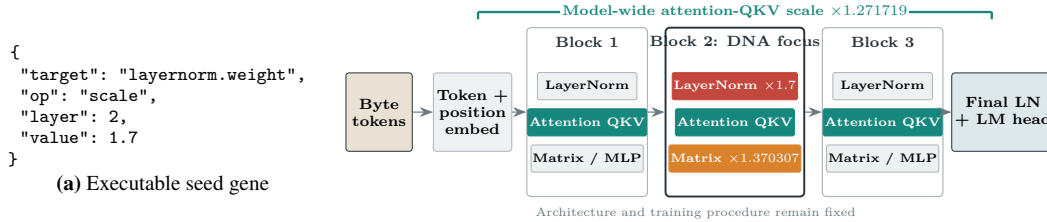


Figure 2: The seed scales layer-2 LayerNorm by 1.7; its child adds layer-2 matrix scaling by 1.370307; and the selected DNA adds model-wide attention-QKV scaling by 1.271719. Over 30 paired seeds, the complete DNA reduces mean steps to PPL 7 from 1465.3 to 931.2 (−36.5%), final PPL from 6.697 to 5.965, and AUC/step from 8.898 to 8.046; the advantage persists through 5000 steps over 20 paired seeds.

The first edge is reconstructed as a preselected replay of the historical random proposal, not as a second blind rediscovery. Search seed 11 is disjoint from validation seeds 22, 33, and 44. The replay reduces their paired mean interpolated steps to perplexity 7 from 1543.3 to 1071.6 (30.56%); the per-seed reductions are 27.39%, 32.12%, and 31.93%. Every run reaches the threshold within 1700 steps.

Table 1: Thirty-seed confirmation. Brackets give 95% confidence intervals for steps to PPL 7; lower is better.

Initialization	Steps to PPL 7 [95% CI]	Final PPL	AUC/step
STANDARD	1465.3 [1447.6, 1482.9]	6.697	8.898
SEED-DNA	1053.1 [1039.8, 1067.0]	6.107	8.290
SELECTED-DNA	931.2 [918.4, 944.1]	5.965	8.046

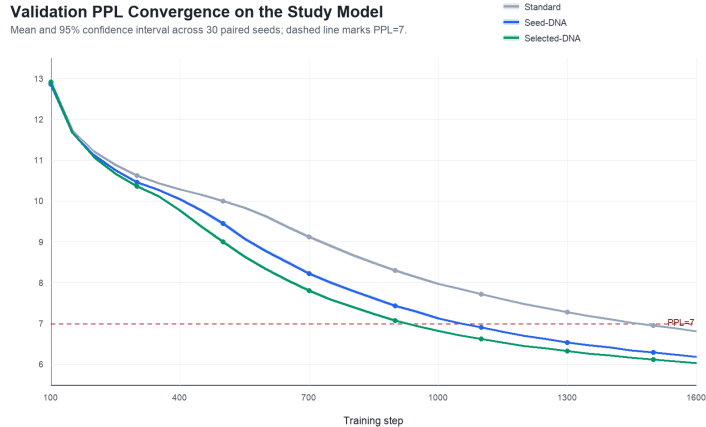


Figure 3: Mean validation-PPL convergence with 95% confidence bands over the same 30 paired confirmatory seeds used in Table 1. Both DNA groups separate from standard initialization early; SELECTED-DNA reaches PPL 7 first and remains lowest through step 1600.

55 The primary metric is the first linearly interpolated step reaching validation PPL 7. Final PPL and
56 validation-loss AUC normalized by steps detect transient crossings. Main confirmation uses 30
57 paired seeds, while long-horizon confirmation uses 20. Mutation fitness is not treated as a confirma-
58 tory estimate: all reported suites evaluate frozen DNA artifacts.

59 3 Controlled Results

60 **Thirty-seed confirmation.** Table 1 gives the primary result. SELECTED-DNA reduces threshold
61 time by 36.5% relative to standard initialization and by 11.6% relative to SEED-DNA. Its final PPL
62 and AUC/step are also lowest, so the gain is not only an earlier threshold crossing.

63 **Learning-rate composition.** The LR-only run confirms that optimizer tuning is powerful: raising
64 the standard group’s rate from 3×10^{-4} to 2×10^{-3} reduces mean threshold time from 1464.2 to
65 507.9 steps. The joint LR+DNA suite evaluates all three artifacts at six rates. At every rate, the
66 mean ordering is SELECTED-DNA < SEED-DNA < STANDARD. The selected DNA’s relative
67 gain attenuates from 35.9% at the default rate to 5.7% at 2×10^{-3} , where it reaches the threshold in
68 478.8 [470.9, 487.7] steps versus 507.9 [494.2, 528.6] for standard initialization. At that strongest
69 rate, SEED-DNA alone is indistinguishable from standard by the reported intervals. Thus the three-
70 gene artifact composes with, rather than replaces, learning-rate tuning.

71 **Component and scale diagnostics.** Removing the matrix and attention genes gives the SEED-
72 DNA row in Table 1, directly quantifying their joint contribution. A separate single-seed LayerNorm
73 family screen varies global scale from 1.3 to 2.0 and reduces threshold time from 1097.9 to 756.4
74 steps. This supports a structured scale neighborhood but does not identify one causal mechanism;
75 the final DNA spans normalization, matrix, and attention groups.

76 **Long-horizon persistence.** Over 20 paired runs of 5000 steps, standard, seed, and selected initial-
77 ization reach PPL 7 in 1467.7, 1055.8, and 935.8 steps, respectively. Their step-5000 PPL values are

78 5.546, 5.320, and 5.275, and AUC/step values are 6.935, 6.495, and 6.362. Both DNA groups beat
 79 their paired standard runs in all 20 seeds at steps 800, 1200, 1600, and 5000; the two-sided sign-test
 80 probability is approximately 1.9×10^{-6} at each checkpoint. The advantage therefore persists beyond
 81 early optimization.

82 4 Interpretation, Scope, and Conclusion

83 The combined evidence supports a focused interpretation. Structured initialization edits alter the
 84 entire training trajectory of the study model, and selection across parameter families yields an artifact
 85 that remains useful after strong optimizer tuning. Weight-space projections and log-count evolution
 86 motivate the view that initialization distributions carry organization relevant to learning speed. They
 87 do not prove that a unique target distribution exists or isolate how normalization, attention, residual
 88 dynamics, and the optimizer interact.

89 The scope is one compact GPT and one mixed corpus. We do not claim transfer to another depth,
 90 width, tokenizer, normalization scheme, or model family. The first mutation is a targeted replay of a
 91 historically random proposal, so it validates that edge on the study model but does not estimate blind
 92 rediscovery probability. Within these limits, initialization DNA provides a compact and auditable
 93 inductive bias: three deterministic scales reduce 30-seed time to target by 36.5%, retain an additional
 94 gain across six learning rates, and improve both time-to-target and fixed-budget quality over 5000
 95 steps.

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